Andrés Araque

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Education

University of Central Florida — Expected 2027

B.S. Mathematics

Coursework: Mathematical Foundations of AI & Machine Learning; Optimization Theory & Mathematical Modeling; Linear Algebra; CS1.

Projects

Hacky / AI matchmaking — ~32 hours · Knight Hacks IX

- Hackathon companion for hacker/mentor matches, events, sponsors, and deadlines.
- Built the React frontend and conversational onboarding; owned branding and messaging.

  Stack: React, Vite, TypeScript, Gemini, Vercel AI SDK, Hono, Neon Postgres.

Étude / Adaptive piano tutor — ~44 hours · Gator Hack V finalist

- Piano tutor scheduling theory and practice, learning and review, on a real 88-key piano.
- Built the learning architecture: 207-skill mastery graph, spaced review, validated planner.

  Stack: TypeScript, React, Next.js, Supabase, Web MIDI.

Donna / Campus connections — On the App Store

- Build UI and backend logic on the founding development team of a UCF social app.
- Lead the agent-first refactor, giving coding agents useful tools and relevant context.

How I Build

seeko-stack - My stack for agentic development.

- Define outcomes; parallelize independent tasks with focused context and isolated worktrees.
- Supervise agents, verify results, preserve task state, and prevent recurring failures.

Work & Teaching

Ultralearn / Stateful AI tutoring — In development

- Building a stateful knowledge-graph tutor for any domain, starting with SAT math.
- Creating content for @ultralearns through content-factory, an agent-driven pipeline that learns from its performance.

  Stack: TypeScript, React, Electron, Next.js, Supabase. **Content pipeline:** Python, Remotion.

A3 Tutors Visit website — Founder & Tutor · 2024-present

I coach and tutor students for the SAT, helping them qualify for Bright Futures scholarships.

Hackathon times are elapsed repository windows for three-person teams, rounded. github.com/seeko-codes